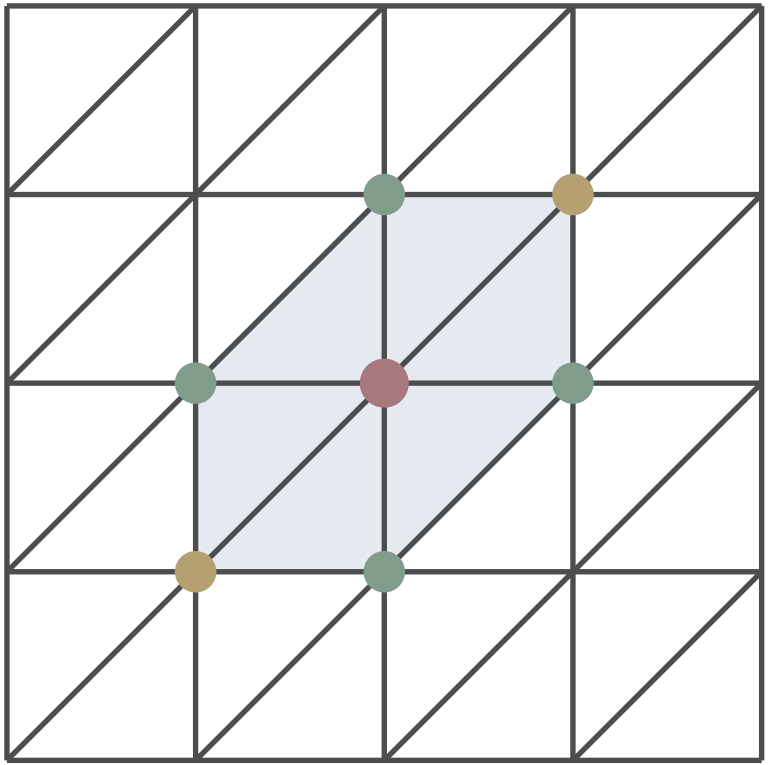
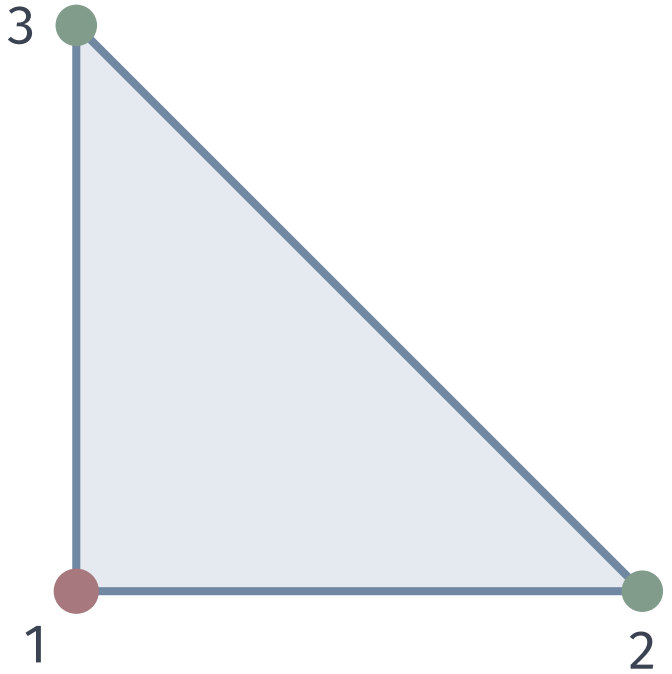


(a)



$\nabla\varphi_1 = (-1,-1)/h$

$\nabla\varphi_2 = (1,0)/h$

$\nabla\varphi_3 = (0,1)/h$

(b)